



Ganling Zhou

917-856-1527 ganlingzhou@outlook.com New York, New York

Education

New York University New York, New York September 2022- May 2026
Bachelor of Music in Music Technology Minors: Mathematics & Computer Science GPA: 3.6/4.0

Skills

Computation & Software: C/C++, Python, Java, Max/MSP, MATLAB, Linux, Microsoft Excel

Hardware & Engineering: CAD Modeling, PCB Design, Hardware Prototyping, Circuit Analysis, Soldering, Embedded Systems (Arduino/Physical Computing)

Signal Processing & Acoustics: Digital Signal Processing (DSP), Audio Signal Routing, Multi-channel Audio Systems, Acoustic Analysis

Languages: English (Fluent), Mandarin (Fluent), French (B2 - Professional Working Proficiency)

Work Experience

Hardware & System Intern, Harvestworks | New York, NY | May 2025 - Present

- Architect and implement complex multi-channel audio/video signal routing and interactive hardware installations for large-scale time-based exhibitions.
- Perform on-site system integration, troubleshooting digital audio networks, embedded systems, and hardware-software communication interfaces.
- Collaborate with developers and creators to optimize interactive sensory hardware and real-time media playback systems.

Technical Assistant & Student Worker, Tisch Production Center, NYU | New York, NY | Aug 2023 - May 2025

- Maintained, tested, and repaired hardware production devices, utilizing precision soldering for cables, connectors, and communication headsets.
- Optimized and audited a large-scale technical asset tracking system via advanced Excel data management, significantly improving equipment deployment efficiency for over 500 active users.
- Analyzed and resolved hardware faults under strict deadlines, maintaining the operational integrity of high-end media capture systems.

Studio Technical Specialist, NYU Steinhardt Media Lab | New York, NY | Jan 2023 - May 2023

- Provided technical infrastructure support and real-time IT/AV troubleshooting for professional recording studios and hybrid computing classrooms.
- Analyzed and drafted comprehensive signal-flow and systemic workflow diagrams to streamline hardware access and diagnostics for user groups.

Projects

THE GUMMY BEAR MACHINE (TANGIBLE DRUM INTERFACE) | EMBEDDED SYSTEMS & HARDWARE

DEVELOPER | 2025 [HTTPS://GANLINGZ.GITHUB.IO/STARWEB/GUMMY-MACHINE.HTML](https://ganlingz.github.io/starweb/gummy-machine.html)

- Exhibition: Officially selected and exhibited at Yale University's CCAM (Center for Collaborative Arts and Media) Fest Fluxus (February 2026).
- Developed an 8-bit, 8-channel physical drum sequencer powered by a **Teensy microcontroller**, mapping tangible UI interactions to real-time multi-track audio synthesis and sample manipulation.
- Engineered a proprietary modular hot-swap mechanism using **magnetic pogo-pin connectors** embedded inside 3D-printed enclosures, utilizing unique custom-soldered resistors to identify and swap entire audio sample sets dynamically via **Teensy analogRead voltage division**.

INTERACTIVE HARDWARE & SMART WEARABLES PROTOTYPING | EMBEDDED SYSTEMS & E-TEXTILE

DEVELOPER | 2026 [HTTPS://GANLINGZ.GITHUB.IO/STARWEB/TUTUTOUCH.HTML](https://ganlingz.github.io/starweb/tututouch.html)

- Designed and built custom PCBs and CAD enclosures for interactive physical computing projects.
- Implemented real-time sensor data mapping using microcontrollers to control dynamic software parameters via serial communication.